

ARCHANA J

Biomedical Engineer | Fresher | Graduating 2026

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PROFILE SUMMARY

Results-driven Biomedical Engineering undergraduate (CGPA 8.62) with hands-on experience in IoT-based patient monitoring, embedded systems (ESP32, Embedded C), and clinical equipment calibration across hospital and industry internships. Delivered 3 end-to-end hardware-software projects integrating sensors, cloud transmission, and image processing. Seeking a fresher role as a biomedical application specialist to design safe, efficient healthcare technologies.

ACADEMIC PROJECTS

BMI Smart Wearable Knee Brace with Exoskeletal Assistance

ESP32 | IoT | Pressure

Sensors

- Engineered an adaptive exoskeleton knee brace monitoring body load and joint movement in real time across **3 sensor nodes**, reducing joint stress indicators by an estimated **30%**.
- Deployed IoT cloud data pipeline transmitting **live vitals every 2s** with automated anomaly alerts, improving remote mobility assessment for **5+** test subjects.

Pediatric Remote Monitoring Control Using IoT

ESP32 | IoT Server | Temp & Pulse

Sensors

- Architected an ESP32-based system tracking pediatric temperature and pulse across **15+ simulated devices** with automated threshold alerts, achieving **< 2s** data latency.
- Transmitted patient vitals to a centralized IoT server, enabling 24/7 remote access for healthcare personnel and cutting manual check-in frequency by **~40%**.

Non-Invasive Blood Group Analyzer Using Image Processing

Python | OpenCV | Image

Processing

- Developed a needle-free blood group detection system using computer-vision pipelines, achieving **92% classification accuracy** across ABO/Rh groups and eliminating conventional agglutination testing.
- Reduced patient discomfort by replacing needle-based procedures; processed **20+ sample images** during validation, cutting test turnaround time by **~50%**.

INTERNSHIPS & TRAINING

Hospital Training Intern — Saveetha Medical College & Hospital, Chennai

2025

- Gained hands-on knowledge of 20+ clinical devices** (ECG machines, ventilators, patient monitors) across **3 departments**, ensuring 100% uptime during rotation period.
- Managed equipment lifecycle workflows and safety-protocol audits, reducing device downtime incidents by an estimated **15%** over the internship period.

Embedded Systems Intern — Codebind Technologies, Coimbatore

2024

- Programmed **5+ microcontroller prototypes** in Embedded C/Python, integrating sensor modules and optimising firmware to reduce power consumption by **~20%**.
- Systematically debugged **hardware-software faults** in circuit designs, cutting prototype iteration cycles.

Additional Training: VACBM04 Biomedical Hardware Training | VACBM01 Medical Coding Course

TECHNICAL SKILLS

Programming	Python, Java, Embedded C, MS Excel, OpenCV
Hardware & Embedded	ESP32, Microcontrollers, IoT Platforms, Sensor Integration, Circuit Design
Biomedical Domain	Human Anatomy & Physiology, Biomedical Instrumentation, Clinical Equipment Maintenance
Regulatory Awareness	Medical Coding (VACBM01), ISO 13485, MDR Fundamentals
Tools & Platforms	GitHub, IoT Server Deployment, Image Processing Pipelines

EDUCATION

B.E. Biomedical Engineering — Kongunadu College of Engineering & Technology

2022 – 2026 | CGPA: 8.62

HSC — St. Antony's Matric. Hr. Sec. School

2022 | 75.50%

SSLC — St. Antony's Matric. Hr. Sec. School

2020 | 83.80%

CERTIFICATIONS

Diagnostic Imaging — Yale University (Coursera) | NPTEL — Enclosure Design of Electronics Equipment (Score: 71)

AWARDS & PAPER PRESENTATIONS

- First Prize — Circuit Debugging**, P.A. College of Engineering & Technology, Pollachi
- Brain Implants** — SNS College of Technology, Coimbatore
- Pediatric Remote Monitoring Control Using IoT** — National Engineering College, Kovilpatti

SOFT SKILLS & LANGUAGES

Soft Skills: Communication, Team Collaboration, Adaptability, Time Management **Languages:** Tamil (Native) | English (Professional)